Task A Baseline Evaluation

We evaluated Qwen3-1.7B without training or modifying its pretrained weights on the course's mixed QA, passage-based RAG, and mathematics tasks. On all 1,007 test examples, the frozen baseline achieved QA exact match of 10.30%, RAG exact match of 37.43%, and mathematical numerical accuracy of 14.68%. The experiment establishes the baseline for the group's subsequent Task B1 comparisons.

1 Model Deployment and Execution

The public checkpoint Qwen/Qwen3-1.7B was deployed in Google Colab Enterprise on an NVIDIA Tesla T4 with 14.56 GiB of GPU memory. Both model and tokenizer were pinned to revision 70d244cc86ccca08cf5af4e1e306ecf908b1ad5e. We used FP16 parameters on cuda:0, SDPA attention, no quantization, no adapter, and no in-context demonstrations.

Setting	Value
Software	PyTorch 2.11.0+cu130; CUDA build 13.0; Transformers 5.20.2
Supporting libraries	Accelerate 1.15.0; Hugging Face Hub 1.33.0; Tokenizers 0.23.3; Safetensors 0.8.0
Decoding	Greedy; enable_thinking=False; batch size 1; seed 5494
Length limits	3,840 input tokens; 256 new output tokens; total budget 4,096
Input handling	Left-padding configuration; no input truncation; reject over-budget inputs

Table 1. Fixed inference settings used for the final-test baseline.

The 9,067 supplied training examples were divided into 8,160 training and 907 validation examples using a fixed, group-aware stratified split with seed 5494. Task A used validation for development; it performed no parameter training. Final evaluation used the separate 1,007 inputs from test.csv. Only Sample_id, Task, Question, and Context entered inference, and Sample_id was retained for alignment with references and predictions.

Prompts were task-specific. QA requested the shortest sufficient answer phrase. RAG required facts supported only by the supplied passages, otherwise the exact response UNANSWERABLE; no additional retrieval system was used. Math requested one number or fraction, without calculations, units, or commentary, and percentage answers as percentage-point numbers. The user templates labeled inputs as Question, Passages plus Question, or Problem, respectively.

The pinned tokenizer's chat template formatted the system and user messages with add_generation_prompt=True and enable_thinking=False. Tokenization then used add_special_tokens=False to avoid duplicating template tokens. Batch size 1 avoided padding between examples, although the tokenizer was configured for left padding. Decoding retained only newly generated tokens and removed special tokens before answer extraction.

Execution and test isolation

After validation, we froze the model, prompts, decoding, and evaluation_v2 rules before running the test set. Stage one generated and saved every prediction before stage two accessed test_answer.csv. The final run recorded zero reference reads during inference, no blank raw responses, no truncated outputs, and a maximum input length of 655 tokens. All 1,007 outputs ended with an end-of-sequence token.

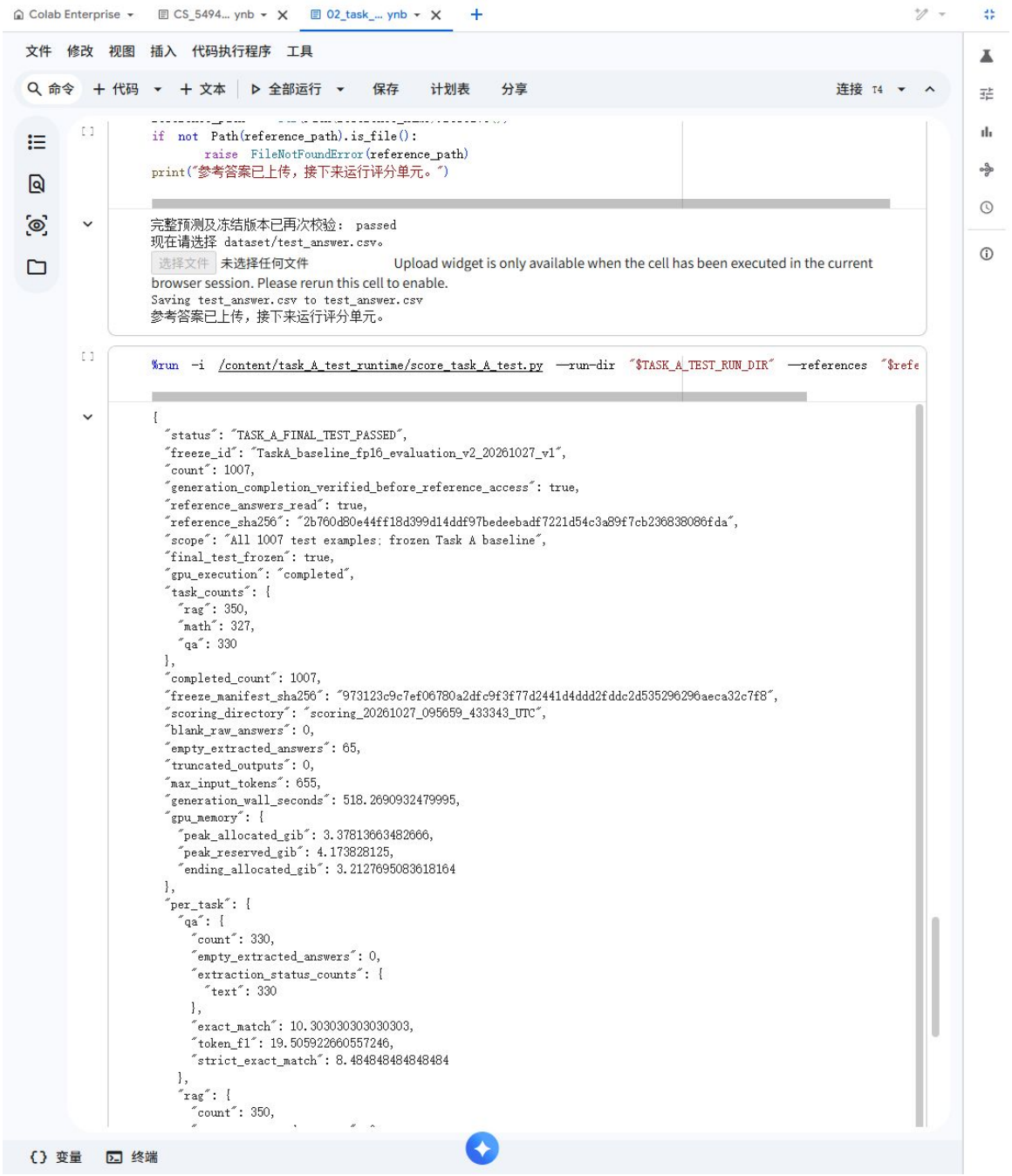


Figure 1. Actual saved Colab Enterprise output from the frozen final-test run on 27 October 2026, captured on 29 October 2026. TASK_A_FINAL_TEST_PASSED, completed_count=1007, and generation_completion_verified_before_reference_access=true confirm completion and the generation-before-scoring sequence. The toolbar reflects the disconnected session at capture time.

The generation loop took 518.27 seconds and peaked at 3.38 GiB of allocated GPU memory (4.17 GiB reserved). This wall-clock measurement includes per-example tokenization, generation, extraction, and logging, and excludes model setup. Saved predictions use the required Sample_id and Answer columns. Re-evaluation of the archived outputs reproduced every per-example score and the reported aggregates.

2 Baseline Results

Metrics and answer processing

For QA and RAG, normalized exact match (EM) is the primary metric because the tasks request concise answers. Normalization applies Unicode NFKC, case folding, and whitespace normalization, while preserving punctuation and English articles. The candidate references are the union of Answer and Accepted_answers. EM and token F1 are independently maximized over those accepted references.

Token F1 measures partial lexical overlap using Unicode word tokens and individual punctuation tokens. It complements EM when an answer contains the reference with extra wording, but does not measure semantic correctness. For RAG items labeled unanswerable, the entire normalized response must equal UNANSWERABLE; an explanatory sentence receives no partial credit for that marker.

Math uses numerical accuracy so that equivalent numbers and fractions can receive the same score. The absolute tolerance is 0.000001, with no relative tolerance. The frozen extractor accepts a complete numeric response or a numeric answer after a recognized terminal final-answer label; it never searches intermediate calculations for a matching value. Percentages follow the prompt's percentage-point convention. Empty, malformed, unparseable, and known-truncated answers receive zero credit and remain in the denominator.

Task	Test n	Correct	Primary metric	Token F1
QA	330	34	EM 10.30%	19.51%
RAG	350	131	EM 37.43%	49.27%
Math	327	48	Numerical accuracy 14.68%	Not applicable

Table 2. Final-test results using all examples and the frozen evaluation_v2 rules. Token F1 is the mean per-example score, expressed as a percentage.

The equal-weight mean of QA EM, RAG EM, and math numerical accuracy is 20.80%. This macro score gives each task equal weight and is not the overall proportion of correct examples. RAG additionally returned the required UNANSWERABLE response for 42 of 60 unanswerable items, giving 70.00% unanswerable accuracy.

Task	Validation n	Validation primary score	Test primary score
QA	300	13.00%	10.30%
RAG	320	34.69%	37.43%
Math	287	15.33%	14.68%
Macro	907 total	21.01%	20.80%

Table 3. Development-validation and final-test results. Both use evaluation_v2, but the splits contain different examples.

During validation, extending the extractor to recognize the terminal labels Final numeric answer and Final numeric answer only increased math accuracy from 6.27% to 15.33% on identical raw outputs. This was an extraction correction, not a model improvement, and was frozen before the final test. The validation-to-test macro difference of -0.20 percentage points is descriptive; it is not evidence of an intervention or statistically established change.

3 Baseline Analysis

General question answering

QA produced 34 exact matches out of 330 examples. A further 76 responses had positive token F1 without an exact match, while 220 had no token overlap with any accepted reference. The low EM and F1 indicate limitations under the fixed short-answer protocol; lexical overlap alone does not establish that an answer is adequate.

For example, TEST_00072 correctly answered yes for the Python expression 'yes' if 4 > 2 else 'no'. In TEST_00004, the response The Revenant did not match the supplied reference Revolutionary Road. These are outcomes under the course's accepted references, rather than an independent verification of every question's wording or factual assumptions.

Passage grounding and abstention

RAG had the highest primary score, with EM 37.43% and token F1 49.27%. Its supplied passages give the model an explicit source for answering, but these results do not isolate a causal benefit of context because the QA and RAG subsets differ in questions and reference conventions. TEST_00022 correctly returned Mrs. Brown from the passage-based question. In TEST_00012, more folk features received EM 0 but token F1 0.80 against folk features, illustrating sensitivity to extra wording.

Sample and outcome	Model output	Reference
TEST_00022 Correct answer	Mrs. Brown	Mrs. Brown
TEST_00012 Extra wording	more folk features	folk features
TEST_00019 Correct abstention	UNANSWERABLE	UNANSWERABLE
TEST_00001 Missed abstention	Sons for the Return Home	UNANSWERABLE
TEST_00003 False abstention	UNANSWERABLE	Mindy Kaling

Table 4. Illustrative RAG outputs. The responses shown are exact model outputs; outcome labels refer to the supplied references.

Abstention errors occurred in both directions. The model correctly abstained on 42 of 60 unanswerable items, but 18 did not produce the required marker. Among the 290 answerable items, 76 responses were exactly UNANSWERABLE, a false-abstention rate of 26.21%; only 89 of those answerable items were exact matches. Thus, the aggregate RAG score includes successful rejection of unsupported questions as well as successful answering.

TEST_00001 answered a question about a novel attributed to Paul Maunder even though the reference required abstention. Conversely, TEST_00003 declined the question about who created the Fox series in which Anna Camp appeared, although the supplied reference was Mindy Kaling. These cases indicate imperfect use of the passage evidence and an imperfect decision about when to answer.

Mathematical computation and output format

Of the 327 math outputs, 48 were numerically correct, 214 were parseable but incorrect, and 65 were unparseable (19.88%). None was truncated, so the extraction failures in this run cannot be explained by reaching the output-token limit. TEST_00018 correctly divided 16 riders among four seats and returned 4. TEST_00002 instead wrote 109 * 74 = 8026 and explicitly finalized 8026, whereas the reference and correct product are 8066. Extraction succeeded, but the computation was wrong.

Sample	Output or exact excerpt	Reference	Score
TEST_00018	4	4	Correct
TEST_00002	109 * 74 = 8026 Final numeric answer only: 8026	8066	Incorrect
TEST_00013	100, 80, 64	64	Unparseable
TEST_00010	Final numeric answer: 13	13	Correct

Table 5. Math examples. TEST_00002 omits only a blank line; TEST_00010 shows the terminal excerpt following a calculation. Other responses are shown in full.

Output format also affected accuracy. TEST_00013 asked for the third anthill's size after two successive 20% decreases from 100 ants. The model returned 100, 80, 64, which contains the reference value but is a list rather than a single final answer. The frozen extractor therefore returned an empty answer and assigned zero credit. In TEST_00010, reasoning was followed by the explicit final label Final numeric answer: 13; the final value was extracted and scored correctly. Disabling the model's thinking mode did not guarantee compliance with the requested answer-only format.

Limitations and implications

The observed weaknesses are low QA exact-answer performance, imperfect grounding and abstention in RAG, and both arithmetic and formatting failures in math. EM can penalize acceptable-looking wording variants that are absent from the reference list, while token F1 can reward overlap in an inadequate answer. Numerical accuracy also reflects instruction following and the declared extraction rules, not only arithmetic ability.

The examples illustrate error types rather than estimate their prevalence. They were drawn from the smallest Sample_id in predefined outcome categories; all reported frequencies were computed over the full test set. The experiment evaluates one frozen model configuration and one test split. It provides no confidence intervals, repeated-seed estimate, or causal attribution to model size, precision, or prompt design.

Task B1 can compare n-shot inference and supervised fine-tuning against these scores using the same test inputs, accepted references, evaluation rules, and matched inference settings. The Task A results alone do not establish whether either extension improves performance. Keeping the baseline fixed makes the subsequent comparison interpretable.

Reproduction artifacts include the two notebooks, the frozen model and evaluation configuration, predictions.csv, raw_outputs.jsonl, and the generation and scoring summaries. The execution entry points and file-upload order are documented in the accompanying README. The final-test run identifier is task_A_test_20261027_094608_541686_UTC.